

Figure 1.30

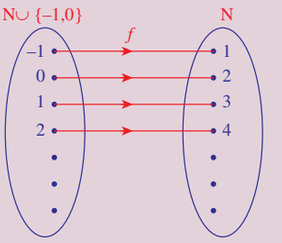


Figure 1.31



It seems that the second function (ii) is same as the first function (i). But the domains are different. From this we see that the domain of the function is also important in deciding whether the function is onto or not. The co-domain has no role in deciding whether the function is one-to-one or not. But it is important to decide whether the function is onto or not.

Example 1.15 Check the following functions for one-to-oneness and ontoness.

(i) $f : \mathbb{N} \to \mathbb{N}$ defined by $f(n) = n^2$.
(ii) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(n) = n^2$.

Solution:

(i) $f(m) = f(n) \Rightarrow m^2 = n^2 \Rightarrow m = n$ since $m, n \in \mathbb{N}$. Thus f is one-to-one. But, non-perfect square elements in the co-domain do not have pre-images and hence not onto.
(ii) Two different elements in the domain have same images and hence f is not one-to-one. Clearly the range of f is a proper subset of $\mathbb{R}$. Thus it is not onto.

Now, we recall Illustration 1.1. In this illustration the function $f : C \to D$ is defined by

$$f(L) = O, f(E) = H, f(T) = W, f(U) = X, f(S) = V, f(W) = Z, f(I) = L, f(N) = Q$$

where $C = \{L, E, T, U, S, W, I, N\}$ and $D = \{O, H, W, X, V, Z, L, Q\}$, is an one-to-one and onto function.

In Illustration 1.3, the function $f : A \to \mathbb{N}$ is defined by $f(a) = 1, f(b) = 2, \ldots, f(z) = 26$, where $A = \{a, b, \ldots z\}$. This function is one-to-one. If we take $\mathbb{N}$ as co-domain, the function is not onto. Instead of $\mathbb{N}$ if we take the co-domain as $\{1, 2, 3 \ldots, 26\}$ then it becomes onto.

Example 1.16 Check whether the following for one-to-oneness and ontoness.

(i) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \dfrac{1}{x}$.

(ii) $f : \mathbb{R} - \{0\} \to \mathbb{R}$ defined by $f(x) = \dfrac{1}{x}$.

Solution:

(i) This is not at all a function because $f(x)$ is not defined for $x = 0$.
(ii) This function is one-to-one (verify) but not onto because 0 has no pre-image.



If we consider $\mathbb{R} - \{0\}$ as the co-domain for the second, then f will become a bijection.